UNIVERSITY OF UDINE
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# Graph-Theoretical Arguments in support of a Quantum Declarative Manifesto

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# Graph Theory for Quantum Computing

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**How?** Through graphs!

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**Is declarative programming the answer?**

# Outline

1 Quantum walks

2 Graph-Theoretical Arguments

3 Declarative Programming for Quantum Computing

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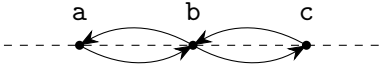

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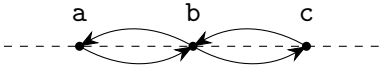


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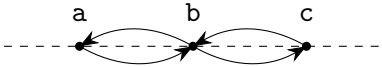


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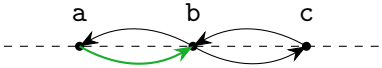


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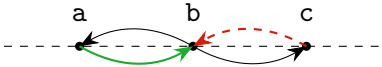


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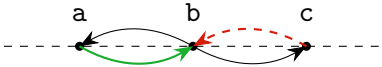


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Edge $(\mathtt{c}, \mathtt{b})$ is lost!

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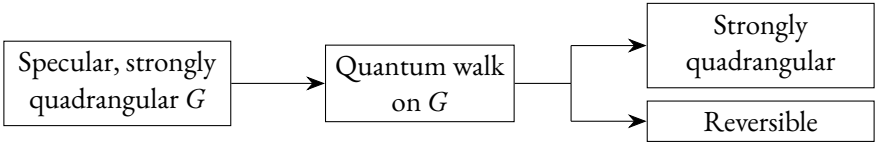
**Question 1.** Which graphs admit quantum walks? [3, 4, 5, 6]



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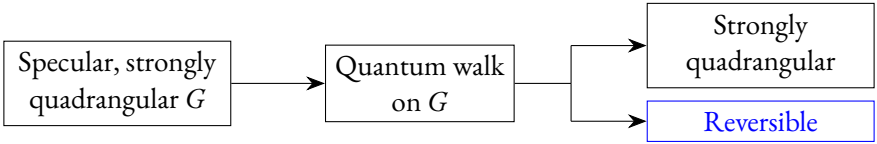


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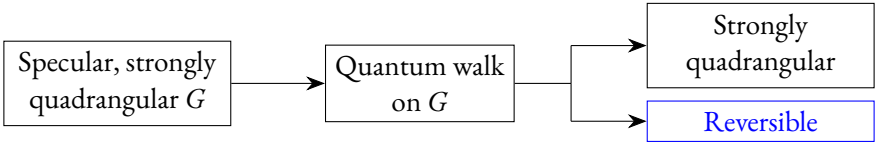


## Reversibility

- Provides a connection to standard graph-theory.
- Almost equivalent to strong connectivity.

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$$\text{Reversibility} \implies \text{Only unitarity is needed} \Downarrow \textbf{Quantum walks}$$

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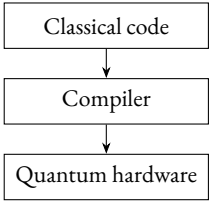

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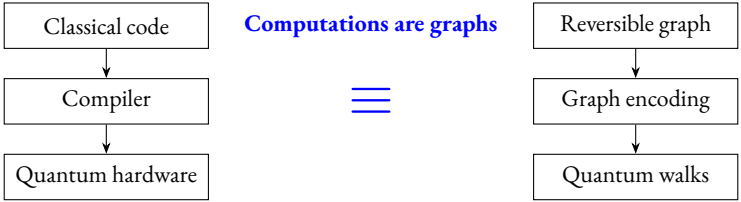

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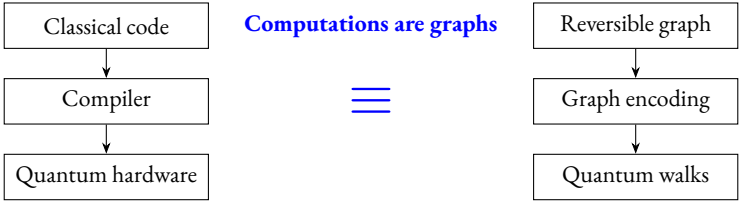


**Which programming paradigm is more suitable for this model?**

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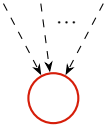

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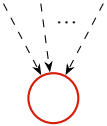


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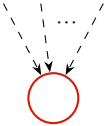


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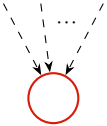


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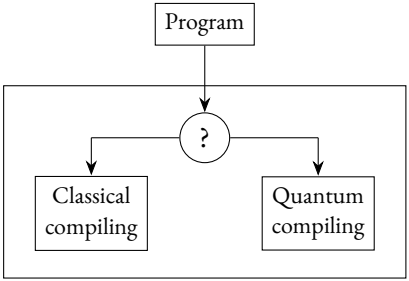

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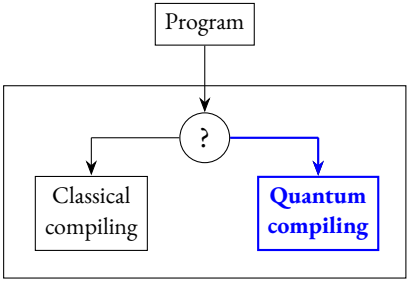

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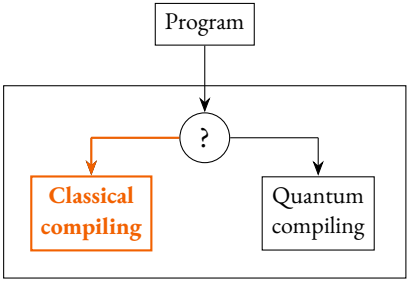

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**SAT solving-based programs would be executable on quantum architecture!**

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Not a promising start. But…

# Efficient storing of graphs

Unitary matrices allow to efficiently store graphs on quantum hardware.

How would this apply to ASP and Minizinc?

## Dependency graphs in ASP

- Tight programs are troubling: acyclic dependency graphs.
- Reversible graphs are characterized by loops.

Not a promising start. But…

These are graphs, not computations. $\Rightarrow$ More liberal encoding approach!

# Conclusions

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No, the common language of Graph Theory will do!

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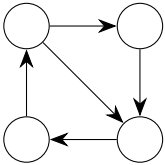

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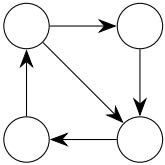

Fig.: Strongly connected graph.

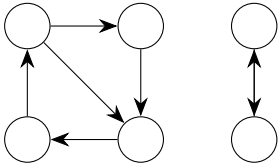

Fig.: Reversible non-strongly connected graph

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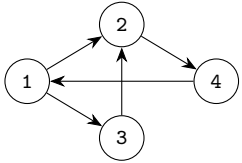


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Fig.: Non-quadrangular graph and its adjacency matrix.

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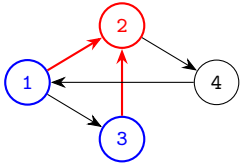


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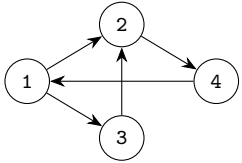


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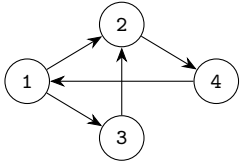


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## Strong quadrangularity

Let $S$ be any set of $k$ vertices, where all $u \in S$ share at least one in- (out-) neighbour with some other $v \in S$. $\Rightarrow$ The union of the pairwise shared in- (out-) neighbours by the vertices in $S$ must be at least $k$.

# Decoding quadrangularity cont'd

- Quadrangular graph not strongly quadrangular.

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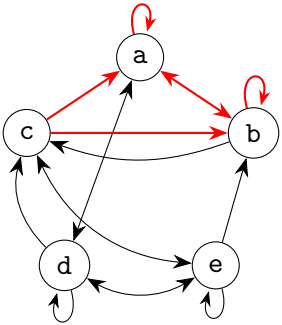
- Quadrangular graph not strongly quadrangular.
- The 3 vertices $\{\mathtt{a},\mathtt{b},\mathtt{c}\}$ share 2 out-neighbours:

$$\{\mathtt{a},\mathtt{b}\}.$$

- $2 < 3$: strong quadrangularity is violated.



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